

Mathematical Formulation of the Proactive Golden-Hour Bridge Model (PGBM) for Post-Earthquake UAV-Assisted Response

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Abstract—This document presents the formal mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM), a UAV-assisted emergency response system for post-earthquake scenarios. The system operates as an online decision-making framework where Responder UAVs are dynamically assigned to detected survivors to provide immediate assistance during the critical "Golden Hour" before ground rescue teams physically arrive. The model captures the coupling between survivor task states, UAV dynamics, battery constraints, time-dependent urgency decay, and 3D routing feasibility. The global system state is formally defined, and the event-triggered decision policy is mathematically characterized.

I. INTRODUCTION

In the aftermath of a major earthquake, Scout UAVs can rapidly detect trapped survivors using thermal and optical sensors. However, Ground Rescue Teams face significant delays due to collapsed infrastructure and blocked roads, creating a deadly gap between detection and physical rescue. The Proactive Golden-Hour Bridge Model (PGBM) introduces an intermediate UAV-assisted response layer where Responder UAVs are dispatched immediately to provide emergency supplies, two-way communication, and GPS relay before ground teams arrive.

A. System Overview

The PGBM consists of four primary actors:

- 1) **Scout UAVs**: Detect survivors and transmit detection packets.
- 2) **Command Center**: Maintains global state and executes online allocation/routing decisions.
- 3) **Responder UAVs**: Execute bridge missions to deliver aid.
- 4) **Ground Rescue Team**: Performs final physical extraction (outside core model).

B. Core Assumptions

- Obstacles are static during a single mission (Version 1).
- Communication between all actors is reliable.
- Each Responder UAV serves one survivor at a time.
- Survivor tasks are independent and non-preemptive (baseline).

II. MATHEMATICAL NOTATION

Throughout this formulation, bold uppercase letters denote sets, bold lowercase letters denote vectors, and scalar values use standard notation. The time variable t denotes the current decision time (continuous). All decisions are event-triggered.

TABLE I
NOTATION SUMMARY

Symbol	Description
$\mathcal{S}(t), \mathcal{D}(t), \mathcal{E}$	Survivor set, Responder UAV set, Environment
$\mathcal{Q}_{\text{high}}, \mathcal{Q}_{\text{low}}$	High/low confidence queues ($c_i \geq c_{\text{thresh}}$)
$S_i = (p_i, t_i^{\text{det}}, c_i, \sigma_i, u_i, a_i)$	Survivor task
$D_j = (p_j, p_j^{\text{base}}, B_j, \text{status}_j)$	Responder UAV
$p_i, p_j, p_j^{\text{base}} \in \mathbb{R}^3$	Positions and base station
$\sigma_i \in [0, 1], c_i \in [0, 1]$	Severity, confidence
$u_i(t) = \sigma_i e^{-\alpha(t-t_i^{\text{det}})}$	Time-decayed urgency, $\alpha = \ln 2 / T_{\text{half}}$
$B_j(t) \in [0, B_{\text{max}}], B_{\text{max}}$	Remaining/max battery
$A_j \in \{0, 1\}, \text{status}_j$	Availability (derived from Idle)
$T_{ij}, E_{ij}^{\text{total}}, E_{\text{return}}$	Travel time, total/return energy
$C_{ij}(t), w_T, w_E, w_U$	Assignment cost and weights ($w_T + w_E + w_U = 1$)
$T_{\text{max}}, E_{\text{max}}$	Normalization constants
$x_{ij}(t) \in \{0, 1\}$	Assignment variable
$X(t) = (\mathcal{S}, \mathcal{D}, \mathcal{E}, \mathcal{Q})$	Global state

III. SYSTEM ENTITIES AND STATE DEFINITIONS

A. Survivor Task Model

Each survivor detection is modeled as a task $i \in \mathcal{S}(t)$, where $\mathcal{S}(t)$ is the set of all known survivor tasks at time t . Survivor i is defined as:

$$S_i = (x_i, y_i, z_i, t_i^{\text{det}}, c_i, \sigma_i, \mathcal{R}_i, a_i, \text{assigned_to}_i) \quad (1)$$

with the following parameters:

$\mathcal{R}_i$ is the set of relief items required by i (e.g., {med}, {med, water}) provided by the Scout/Command Center; $|\mathcal{R}_i| \geq 1$ and $|\mathcal{R}_i| > 1$ denotes multiple demands at the same location. Severity σ_i denotes the static medical criticality at detection (independent of time), with $\sigma_i = 1$ representing the most critical condition. Two possible scenarios for obtaining σ_i are:

TABLE II
SURVIVOR TASK PARAMETERS

Parameter	Symbol	Domain
Location	(x_i, y_i, z_i)	$\mathbb{R}^3$
Detection Time	t_i^{det}	$\mathbb{R}^+$
Confidence Score	c_i	$[0, 1]$
Severity	σ_i	$[0, 1]$
Demand Set	$\mathcal{R}_i$	$\subseteq \{\text{med, water, food}\}$
Assignment Flag	a_i	$\{0, 1\}$
Assigned UAV	assigned_to _{i}	$\mathcal{D}(t) \cup \{\emptyset\}$

(1) *Scout AI Inference*: estimated automatically from Scout UAV thermal/optical imagery and rubble context via a classification model (e.g., motion/heat signature analysis), and (2) *Human-in-the-Loop*: tagged manually by the Command Center operator reviewing the Scout video feed. If no severity estimate is available, $\sigma_i = 1$ uniformly.

B. Urgency Decay Function

The urgency of survivor i decays exponentially with waiting time according to the Golden Hour principle:

$$u_i(t) = \sigma_i \cdot e^{-\alpha(t-t_i^{\text{det}})} \quad (2)$$

where $\sigma_i \in [0, 1]$ is the severity at detection so that $u_i(t_i^{\text{det}}) = \sigma_i$, $\Delta t = t - t_i^{\text{det}} \geq 0$ is the waiting time, and $\alpha > 0$ is the decay rate. The rate is defined via the half-life T_{half} (e.g., $T_{\text{half}} = 30$ min) as

$$\alpha = \frac{\ln 2}{T_{\text{half}}}, \quad (3)$$

so that $u_i(t_i^{\text{det}} + T_{\text{half}}) = \sigma_i/2$. For $T_{\text{half}} = 30$ min, $\alpha \approx 0.023 \text{ min}^{-1}$ ($3.8 \times 10^{-4} \text{ s}^{-1}$); $T_{\text{half}} = 60$ min gives $\alpha \approx 0.0115 \text{ min}^{-1}$. A survivor found 10 min ago ($\Delta t = 10$) has $u \approx 0.79\sigma_i$, while one found 90 min ago has $u \approx 0.12\sigma_i$ at the same current time t , capturing higher salvageability of fresher detections.

C. Confidence Score and Verification Policy

Each detection carries a confidence score $c_i \in [0, 1]$ representing the Scout UAV detector's estimated probability that the detection is a true survivor (e.g., thermal/optical AI output). A threshold $c_{\text{thresh}} \in [0, 1]$ (e.g., 0.5) separates high- and low-confidence regimes:

$$\text{eligible}(i) = \begin{cases} 1 & \text{if } c_i \geq c_{\text{thresh}} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

Detections with $c_i \geq c_{\text{thresh}}$ enter the high-confidence queue $\mathcal{Q}_{\text{high}}(t)$ and are eligible for Responder assignment. Detections with $c_i < c_{\text{thresh}}$ enter the low-confidence queue $\mathcal{Q}_{\text{low}}(t)$ and are *not* eligible for Responder dispatch; they require verification (Scout re-scan or operator review) before promotion to $\mathcal{Q}_{\text{high}}$. In Version 1, $\mathcal{Q}_{\text{low}}$ is modeled as a holding set and verification is abstracted as an exogenous promotion event. The offloading ILP is defined only over $\mathcal{Q}_{\text{high}}(t)$.

D. Responder UAV Model

Each Responder UAV $j \in \mathcal{D}(t)$ is defined as:

$$D_j = (x_j, y_j, z_j, p_j^{\text{base}}, B_j, K, I_j, m, A_j, s_j, \text{status}_j) \quad (5)$$

with the following parameters:

TABLE III
RESPONDER UAV PARAMETERS

Parameter	Symbol	Domain
Position	(x_j, y_j, z_j)	$\mathbb{R}^3$
Base Station	p_j^{base}	$\mathbb{R}^3$
Remaining Battery	B_j	$\mathbb{R}^+$
Capacity	K	$\mathbb{N}$ (e.g., 4 items)
Inventory	I_j	multiset $\subseteq \{\text{med, water, food}\}$
Payload Mass	m	$\mathbb{R}^+$ (kg, $m = \sum_{\text{item} \in I_j} \text{weight}(\text{item})$)
Availability	A_j	$\{0, 1\}$
Current Task	s_j	$\mathcal{S}(t) \cup \{\emptyset\}$
Mission Status	status_j	$\{\text{Idle, EnRoute, AtSurvivor, Returning}\}$

K is the maximum number of items, I_j is the loaded inventory (e.g., $[\text{med, water, food, med}]$), and m is the derived total weight of I_j which decreases after each drop and affects e_{travel} per leg.

E. Availability States

The availability A_j is derived from the UAV's status:

$$A_j = \begin{cases} 1 & \text{if } \text{status}_j = \text{Idle} \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

F. Environment Model

The disaster environment is represented as continuous sets:

$$\mathcal{E} = (V, O) \quad (7)$$

where $V \subset \mathbb{R}^3$ is navigable free space and $O \subset \mathbb{R}^3$ is the static obstacle set (buildings, rubble, no-fly zones), with $V \cap O = \emptyset$. For Version 1, $O(t) = O$ (static) and the hazard map $R : V \rightarrow [0, 1]$ is set to $R = 0$ and deferred. A route P is feasible iff $P \subset V$ (all waypoints lie in free space). No discretization or algorithm is assumed at formulation level.

G. Task Queue

The ordered queue of unassigned survivors is defined as a derived view:

$$\mathcal{Q}(t) = \text{order}(\{S_i \in \mathcal{S}(t) \mid a_i = 0\}) \quad (8)$$

ordered by priority:

$$\text{priority}(i) = \beta_1 \cdot u_i(t) + \beta_2 \cdot c_i - \beta_3 \cdot (t - t_i^{\text{det}}) \quad (9)$$

Note: $\mathcal{Q}(t)$ is not an independent state component but a derived view of $\mathcal{S}(t)$ for scheduling. With verification threshold c_{thresh} , the queue partitions as:

$$\mathcal{Q}_{\text{high}}(t) = \{S_i \in \mathcal{Q}(t) \mid c_i \geq c_{\text{thresh}}\}, \quad (10)$$

$$\mathcal{Q}_{\text{low}}(t) = \{S_i \in \mathcal{Q}(t) \mid c_i < c_{\text{thresh}}\} \quad (11)$$

Only $\mathcal{Q}_{\text{high}}(t)$ is eligible for the offloading assignment; $\mathcal{Q}_{\text{low}}(t)$ is held for verification. *Uncertainty handling:* The partition manages detection uncertainty (false positives from Scout thermal/optical AI). High-confidence detections are directly eligible, while low-confidence detections are buffered in $\mathcal{Q}_{\text{low}}$ for verification (re-scan/operator review) before promotion, preventing scarce Responder UAVs from being wasted on likely false alarms. Note: $\text{priority}(i)$ is a pre-filter ordering heuristic for $\mathcal{Q}_{\text{high}}(t)$ (e.g., for greedy fallback or tie-breaking); the optimal assignment is determined by the ILP cost $C_{ij}(t)$, not by priority order.

IV. DECISION VARIABLES

A. Assignment Variable

$$x_{ij}(t) \in \{0, 1\} \quad (12)$$

$$x_{ij}(t) = \begin{cases} 1 & \text{if UAV } j \text{ is assigned to survivor } i \text{ at time } t \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

B. Route Variable (Multi-Drop Tour)

For multi-drop, the decision is a tour $\pi_j = (i_1, i_2, \dots, i_k)$ with $k \leq K$ ordered survivors, where $|\pi_j| = \sum_{i \in \pi_j} |\mathcal{R}_i| \leq K$. The tour route is the concatenation

$$P_{\pi_j} = P_{p_j \rightarrow p_{i_1}} \circ P_{p_{i_1} \rightarrow p_{i_2}} \circ \dots \circ P_{p_{i_k} \rightarrow p_j^{\text{base}}} \quad (14)$$

where $\circ$ denotes concatenation and each leg $P_{a \rightarrow b} = \{a = p_0, p_1, \dots, p_K = b\}$ is itself a waypoint sequence with $p_k \in V$ (e.g., $P_{p_j \rightarrow i_1}$ may be $\{p_j, x_1, x_2, p_{i_1}\}$). The tour is obtained from the routing oracle: each leg is queried as $\text{RoutingOracle}(p_a, p_b)$, which returns (T_{ab}, E_{ab}) for the optimal leg or (∞, ∞) if no $P_{a \rightarrow b} \subset V$ exists. Thus P_{π_j} is the concatenation of $k+1$ leg paths, and $T(\pi_j), E(\pi_j)$ are the sums of the leg T and E .

V. COST FUNCTIONS

A. Path Cost

The generic cost of a path $P = \{p_0, \dots, p_K\}$ is:

$$\text{Cost}(P) = \sum_{k=1}^K (w_d \|p_k - p_{k-1}\| + w_z \max(0, z_k - z_{k-1})) \quad (15)$$

where $w_d, w_z > 0$ weight horizontal distance and vertical climb.

B. Optimal Path

The cheapest feasible path from p_j to p_i is

$$P_{ij}^* = \arg \min_{P \subset V, p_0=p_j, p_K=p_i} \text{Cost}(P) \quad (16)$$

where the minimum is taken over all paths lying entirely in V . If no such path exists, P_{ij}^* is infeasible and the pair (i, j) is unreachable. When feasible, the travel time T_{ij} and energy E_{ij} are both derived from this single path P_{ij}^* . The routing oracle encapsulates this: given any two points it returns the corresponding T , E , or infinity if no feasible path exists. Offloading queries the oracle twice, for the outbound leg $p_j \rightarrow p_i$ and the return leg $p_i \rightarrow p_j^{\text{base}}$; both must be feasible for the assignment to be feasible.

C. Travel Time per Tour

For a tour $\pi_j = (i_1, \dots, i_k)$, the total travel time is the sum over legs:

$$T(\pi_j) = \sum_{l=0}^k T_{\text{leg}}(l), \quad T_{\text{leg}}(l) = \frac{\|P_l\|}{v_{\text{max}}} \quad (17)$$

where $P_0 = P_{p_j \rightarrow p_{i_1}}^*$, $P_l = P_{p_{i_l} \rightarrow p_{i_{l+1}}}^*$ for $1 \leq l < k$, $P_k = P_{p_{i_k} \rightarrow p_j^{\text{base}}}^*$, v_{max} is the maximum speed, and an additional $T_{\text{hover}} = 15$ s per stop is added for drop confirmation.

D. Energy per Tour

Let $m_0 = \sum_{item \in I_j} \text{weight}(item)$ be the initial payload and m_l the remaining payload before leg l ($m_l = m_{l-1} - \sum_{item \in \mathcal{R}_{i_l}} \text{weight}(item)$). The energy of leg l is

$$E_{\text{leg}}(l) = e_{\text{travel}}(p_l, m_{l-1}, w(p_l)) \|P_l\| + e_{\text{vert}}(m_{l-1}) \Delta z_l \quad (18)$$

with $e_{\text{travel}}(p, m, w) = e_0 + k_m m + k_w w(p)$ and $e_{\text{vert}}(m) = mg/\eta_{\text{vert}}$ as above ($m, w(p_l)$ realtime). The total tour energy is

$$E(\pi_j) = \sum_{l=0}^k E_{\text{leg}}(l) + \sum_{i \in \pi_j} E_{\text{hover}}(i), \quad E_{\text{hover}} = e_{\text{hover}} T_{\text{hover}} \quad (19)$$

with $T_{\text{hover}} = 15$ s per stop. Early legs use larger m and thus higher e_{travel} .

E. Total Energy per Tour

The total energy for tour π_j is $E(\pi_j)$ as defined above; the single-leg E_{ij}^{total} is the special case $k=1$. where $P_{ij}^* = P_{j \rightarrow i}^*$. Variant $E_{ij}^{\text{oneway}} = E(P_{ij}^*) + E_{\text{hover}}$ (landing) is deferred; V1 assumes roundtrip.

F. Coupling to Offloading

Routing provides $(T_{ij}, E_{ij}^{\text{total}})$ via the oracle for each pair (i, j) . Offloading uses these values in the feasibility check (21) and the assignment cost (20). Both the outbound leg $p_j \rightarrow p_i$ and the return leg $p_i \rightarrow p_j^{\text{base}}$ must be feasible; otherwise $C_{ij}(t) = \infty$.

G. Assignment Cost

The cost of assigning UAV j to survivor i is:

$$C_{ij}(t) = \begin{cases} w_T \frac{T_{ij}}{T_{\max}} + w_E \frac{E_{ij}^{\text{total}}}{E_{\max}} - w_U u_i(t) c_i & \text{if Feasible}(i, j) \\ \infty & \text{otherwise} \end{cases} \quad (20)$$

where $w_T, w_E, w_U > 0$ are weighting coefficients (cost weights, distinct from priority weights $\beta_1, \beta_2, \beta_3$; e.g., $w_T + w_E + w_U = 1$ for normalized trade-off), $T_{\max}$ and $E_{\max}$ are normalization constants (maximum feasible time/energy), and the term $-w_U \cdot u_i(t) \cdot c_i$ is the expected urgency reward: higher $u_i \cdot c_i$ reduces cost, incentivizing assignment to more urgent/confident survivors. Infeasible pairs have infinite cost and are excluded from the ILP.

Rationale: The cost is a dimensionless trade-off among (1) response time $w_T T_{ij}/T_{\max}$, (2) energy $w_E E_{ij}^{\text{total}}/E_{\max}$, and (3) expected value $-w_U u_i(t) c_i$ with $u_i(t) = \sigma_i e^{-\alpha(t-t_i^{\text{det}})}$. Normalization makes w_T, w_E, w_U comparable ($w_T + w_E + w_U = 1$ baseline); increasing w_U favors urgent/confident survivors despite distance, while larger w_T, w_E favor nearest, cheapest assignments. ∞ enforces feasibility before optimization.

VI. FEASIBILITY CONSTRAINTS

A. Battery Constraint for Tour

A tour π_j is feasible only if the whole tour plus reserve fits the remaining battery:

$$E(\pi_j) + E_{\text{reserve}} \leq B_j(t) \quad (21)$$

where $E(\pi_j) = \sum_l E_{\text{leg}}(l) + \sum_{i \in \pi_j} E_{\text{hover}}(i)$ from Step 4 and $E_{\text{reserve}} = 0.2 B_{\max}$.

B. Demand Matching

Each demand set must be satisfied from the remaining inventory:

$$\mathcal{R}_{i_l} \subseteq I_j^{(l-1)} \quad \forall i_l \in \pi_j \quad (22)$$

where $I_j^{(l-1)}$ is the inventory before stop l ($I_j^{(0)} = I_j$). If $\mathcal{R}_{i_l} \not\subseteq I_j^{(l-1)}$, the tour is infeasible.

C. Reachability per Leg

Each leg must lie in free space: $P_l \subset V$ for all l , i.e., $p_k \in V$ for every waypoint. Reachability is an oracle per leg; one infeasible leg makes the whole tour infeasible.

D. Capacity and Availability

$|\pi_j| = \sum_{i \in \pi_j} |\mathcal{R}_i| \leq K$ and $A_j = 1$ (Idle). If $A_j = 0$ or capacity exceeded, the tour is infeasible.

E. Feasible Tour Condition

A tour π_j is feasible iff all of the above hold:

$$\begin{aligned} \text{Feasible}(\pi_j) = & (A_j = 1) \wedge (|\pi_j| \leq K) \wedge (\forall i \in \pi_j : c_i \geq c_{\text{thresh}}) \\ & \wedge (\forall l : P_l \subset V) \wedge (\forall i_l \in \pi_j : \mathcal{R}_{i_l} \subseteq I_j^{(l-1)}) \\ & \wedge (E(\pi_j) + E_{\text{reserve}} \leq B_j(t)) \end{aligned} \quad (23)$$

VII. ASSIGNMENT CONSTRAINTS

A. Single Assignment per Survivor

Each survivor in $\mathcal{Q}_{\text{high}}(t)$ appears in at most one tour:

$$\sum_j \mathbf{1}\{i \in \pi_j(t)\} \leq 1 \quad \forall i \in \mathcal{Q}_{\text{high}}(t) \quad (24)$$

B. Capacity per Tour

Each tour respects item capacity:

$$|\pi_j(t)| = \sum_{i \in \pi_j(t)} |\mathcal{R}_i| \leq K \quad \forall j \in \mathcal{D}_{\text{avail}}(t) \quad (25)$$

where K is the UAV capacity (e.g., 4) and $|\mathcal{R}_i|$ is the number of items demanded by i . A UAV may stay idle ($\pi_j = \emptyset$) if no feasible tour exists.

C. Feasibility Enforcement

If a tour π_j is not feasible, its cost is set to infinity: $C(\pi_j) = \infty$ when $\neg \text{Feasible}(\pi_j)$ (includes $c_i < c_{\text{thresh}}$, $A_j = 0$, $P_l \not\subset V$, $\mathcal{R}_{i_l} \not\subseteq I_j^{(l-1)}$, or $E(\pi_j) + E_{\text{reserve}} > B_j$). Any optimal solution will have such tours excluded.

VIII. OBJECTIVE FUNCTION

A. Primary Objective: Minimize Cost

The snapshot offloading problem OFF-SNAP(t) over $\mathcal{Q}_{\text{high}}(t) \times \mathcal{D}_{\text{avail}}(t)$ is:

$$\min_{\mathbf{x}(t)} \sum_{i \in \mathcal{Q}_{\text{high}}(t)} \sum_{j \in \mathcal{D}_{\text{avail}}(t)} x_{ij}(t) \cdot C_{ij}(t) \quad (26)$$

If $\mathcal{Q}_{\text{high}}(t) = \emptyset$ or all $C_{ij} = \infty$, the optimum is $x_{ij} = 0$ (no dispatch, wait for next event).

IX. SCOPE AND LIMITATIONS

A. Core Model Includes

- Survivor detection as exogenous input
- Responder UAV assignment and routing
- Battery-aware feasibility
- Time-dependent urgency
- Online, event-triggered decision making

B. Excluded (Future Work)

- Dynamic aftershock obstacles
- UAV-to-UAV collision avoidance
- Communication relay UAVs
- Hot-swap battery stations
- Detection uncertainty verification policies
- Recourse/reassignment after dispatch
- High-fidelity nonlinear energy/power model (mass, aerodynamic drag, wind, acceleration-aware)

X. CONCLUSION

This document presents a complete mathematical formulation of the Proactive Golden-Hour Bridge Model (PGBM) for post-earthquake UAV-assisted response. The model captures the essential dynamics of the system: survivor states with time-dependent urgency, Responder UAV states with battery constraints, and the online nature of detection arrivals and resource availability. The formulation separates feasibility (battery and routing constraints) from optimization (assignment and path selection), providing a clear foundation for algorithm design and simulation. The coupling between assignment and routing is explicitly defined through the cost function C_{ij} , which depends on both travel time and energy consumption.